



ARTIFICIAL INTELLIGENCE

the policy network decides on the next move, while another network called the value network predicts the winner of the game.

After learning from over 30 million human moves, the system could predict the human move around 57 per cent of the time. Then, AlphaGo learnt to better these human moves by discovering new strategies using a method called reinforcement learning. Basically, the system played innumerable games between its neural networks and adjusted the connections using a trial-and-error process. Google Cloud provided the extensive computing power needed to achieve this.

What made AlphaGo win at a game that baffled computers till then was the fact that it could figure out the moves and winning strategies by itself, instead of relying on hand-crafted rules. This makes it an ideal example of deep learning.

AI world has always used games to prove its mettle, but the same talent can be put to better use. DeepMind is working on systems to tackle problems ranging from climate modelling to disease analysis. Google itself uses a lot of deep learning. According to a statement by Mustafa Suleyman, co-founder of DeepMind, deep learning networks have now replaced 60 handcrafted rule based systems at Google.

Learning from Jeopardy

Some trend-watchers claim that it is Watson, IBM's AI brainchild, which transformed IBM from a hardware company to a business analytics major. Watson was a path-breaking natural language processing (NLP) computer, which could answer questions asked conversationally. In 2011, it made the headlines by beating two champions at the game of Jeopardy. It was immediately signed on by Cleveland Clinic to synthesise humongous amounts of data to generate evidence based hypotheses, so they could help clinicians and students to diagnose diseases more accurately and plan

their treatment better.

Watson is powered by DeepQA, a software architecture for deep content analysis and evidence-based reasoning. Last year, Watson strengthened its deep learning abilities with the acquisition of AlchemyAPI, whose deep learning engines specialise in digging into Big Data to discover important relationships.

In a media report, Steve Gold, vice president of the IBM Watson group, said that, "AlchemyAPI's technology will be used to help augment Watson's ability to identify information hierarchies and understand relationships between people, places and things living in that data. This is particularly useful across long-tail domains or other ontologies that are constantly evolving. The technology will also help give Watson more visual features such as the ability to detect, label and extract details from image data."

IBM is constantly expanding its line of products for deep learning. Using IBM Watson Developer Cloud on Bluemix, anybody can embed Watson's cognitive technologies into their apps and products. There are APIs for NLP, machine learning and deep learning, which could be used for purposes like medical diagnosis, marketing analysis and more.

APIs like Natural Language Classifier, Personality Insights and Trade-off Analytics, for example, can help marketers. Data First's influencer technology platform, Influential, used Watson's Personality Insights API to scan and sift through social media and identify influencers for their client, KIA Motors. The system looked for influencers who had traits like openness to change, artistic interest and achievement-striving. The resulting campaign was a great success.

Quite recently, IBM and Massachusetts Institute of Technology got into a multi-year partnership to improve AI's ability to interpret sight and sound as well as humans. Watson is expected to be a key part of this research. In September, IBM



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